

# MS&E 228: Applied Causal Inference Powered by ML and AI

## Lecture 12: Non-Parametric Structural Equations and Directed Acyclic Graphs

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**Readings:** *Applied Causal Inference Powered by ML and AI*, §7, §11.

## Where We Are

We have completed a long journey through causal machine learning estimation:

- ▶ **Potential outcomes framework:**  $Y(1)$ ,  $Y(0)$ , ATE, ATT, CATE.
- ▶ **Key identification assumption:** Conditional exogeneity  $Y(d) \perp\!\!\!\perp D \mid X$ .
- ▶ **Two identification strategies:** Conditioning & propensities.
- ▶ **Estimation paradigms:** DR/DML estimators; DR-Learner and R-Learner for CATE.
- ▶ **Overarching principle:** “Insensitive estimation formulas” robust to nuisance model errors.

*We emerged victorious from causal ML estimation! Now we switch gears completely.*

# The Big Question

We go back to the basics — the real “Causal Inference.”

## The Motivating Question

The main assumption that enabled identification was **conditional exogeneity**. But how can we *deduce* it from more **human-interpretable** assumptions?

Imagine going to a domain expert and asking:

*“Do you believe that your potential outcomes are conditionally independent of the treatment given the covariates?”*

Most domain experts won't even grasp what this means!

**Our goal:** Find a more **human-friendly** way of expressing assumptions on the data generating process, such that conditional exogeneity is a **conclusion** — not the premise.

# What We Will Achieve

In resolving this issue, we will achieve something far greater:

1. **A new language: Non-parametric structural equation models (SEMs)** — a richer, mechanistic foundation for the potential outcomes framework.
2. **A visual representation: Directed acyclic graphs (DAGs)** — a perfect fit for human-interpretable assumption elicitation.
3. **Interventions formalized:** A formal mathematical concept of an “intervention” that gives potential outcomes an **interventional interpretation**.
4. **Powerful tools:** The **Single World Intervention Graph (SWIG)** to deduce conditional exogeneity and to identify good vs. bad controls.

*This framework is not confined to conditional exogeneity — it will help us derive other identification strategies later on.*

# **Part I: From Domain Knowledge to Causal Graphs**

Building the Influence Graph from an Empirical Example

# The Hospital Messaging Example — Setup

Let's revisit the hospital messaging application from the in-class activity in the last lecture.

## Setting

Doctors monitor young patients with diabetes through a remote monitoring system and send messages to incentivize better blood glucose control.

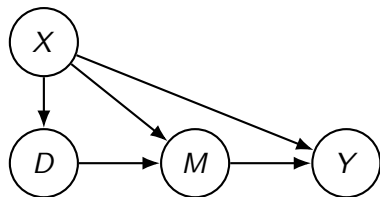
The key variables:

- ▶  $Y$  = glucose time-in-range in the days after the message
- ▶  $D$  = whether the doctor decided to send a message
- ▶  $M$  = the actual text of the message
- ▶  $X$  = all variables the doctor observes in the system (e.g., CGM data)

# The Influence Graph — Observed Variables

Here is one reasonable “influence graph” for the observed variables:

- ▶  $X \rightarrow D$ : CGM data influences the doctor's decision.
- ▶  $D \rightarrow M$ : Decision to send leads to message text.
- ▶  $X \rightarrow M$ : CGM data influences message content.
- ▶  $X \rightarrow Y$ : Patient history influences outcome.
- ▶  $M \rightarrow Y$ : The message influences glucose outcome.



*But are there other variables that influence the observed variables?*

# Adding Unobserved / Latent Variables

Yes! There are unobserved variables that influence the observed ones:

## Patient behavioral habits ( $U_p$ ):

- ▶ What they eat, exercise habits, etc.
- ▶  $U_p \rightarrow X$ : habits influence monitoring data
- ▶  $U_p \rightarrow Y$ : habits influence glucose outcome

## Doctor's capacity / workload ( $U_c$ ):

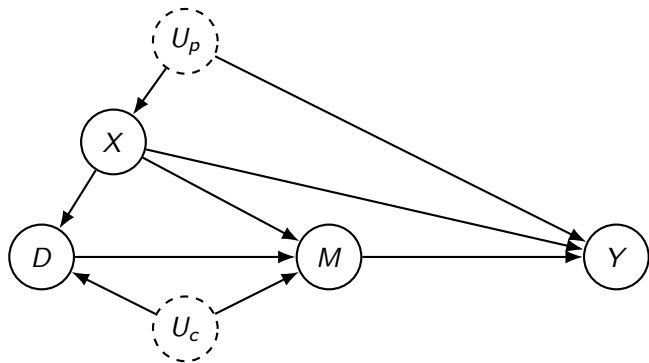
- ▶  $U_c \rightarrow D$ : workload affects decision to send
- ▶  $U_c \rightarrow M$ : workload affects message quality

## Noise terms ( $\varepsilon_X, \varepsilon_D, \varepsilon_M, \varepsilon_Y$ ):

Each only influences its own node. We **omit these from the graph** for clarity and always keep in mind that each node has an associated independent noise term.



# The Complete DAG



This is the **human-interpretable assumption** we are willing to make. This graph describes how we believe, after conversing with a domain expert, the data we observe are generated.

*Dashed circles = unobserved. Each node also has implicit independent noise term  $\varepsilon_i$ .*

# The DAG as a Bayesian Network

This graph is a **Bayesian network** for the data:

## Markov Factorization

The probability law of the data factorizes according to the graph:

$$p(\mathbf{v}) = \prod_i p(v_i \mid v_{\text{Parents}(i)})$$

## What does this mean?

- ▶ Each variable is conditionally independent of its non-descendants given its parents.
- ▶ The graph captures **all conditional independencies** that hold in the data.

*In our example:*

$$p(u_p, u_c, x, d, m, y) = p(u_p) p(u_c) p(x \mid u_p) p(d \mid x, u_c) p(m \mid d, x, u_c) p(y \mid m, x, u_p)$$

# **Part II: Non-Parametric Structural Equations**

What the DAG Really Means Mathematically

# From DAG to Structural Equations

The DAG means: every variable is generated as a **function of its parent variables** and an **exogenous noise term**:

## Non-Parametric Structural Equation Model (SEM)

$$\begin{aligned}X &= f_X(U_p, \varepsilon_X), \\D &= f_D(X, U_c, \varepsilon_D), \\M &= f_M(D, X, U_c, \varepsilon_M), \\Y &= f_Y(M, X, U_p, \varepsilon_Y)\end{aligned}$$

- ▶ **No restriction** on how these functions look like (non-parametric!).
- ▶ The **crucial restriction** is on *what inputs* each function takes — only the **parents** of the node in the graph.
- ▶ Noise terms  $\varepsilon_X, \varepsilon_D, \varepsilon_M, \varepsilon_Y$  and latent factors  $U_p, U_c$  are **jointly independent**.

# The Data Generating Process

The SEM defines a **sequential data generating process**:

1. **Draw** the jointly independent exogenous variables:  
 $U_p, U_c, \varepsilon_X, \varepsilon_D, \varepsilon_M, \varepsilon_Y \sim \text{some exogenous distribution.}$
2. **Generate**  $X \leftarrow f_X(U_p, \varepsilon_X)$
3. **Generate**  $D \leftarrow f_D(X, U_c, \varepsilon_D)$
4. **Generate**  $M \leftarrow f_M(D, X, U_c, \varepsilon_M)$
5. **Generate**  $Y \leftarrow f_Y(M, X, U_p, \varepsilon_Y)$

## Key Insight

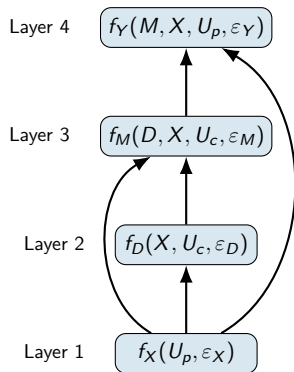
The DAG that we depicted from domain assumptions implicitly defines a **generative model** for the data!

## Side Note: Connection to Deep Generative Models

There is an interesting new line of work that takes this viewpoint and connects it to **deep generative models**:

- ▶ Each structural equation  $f_i$  can be modeled as a **neural network layer**.
- ▶ The SEM becomes a deep generative model.
- ▶ Causal inference reduces to training deep generative models with gradient descent!

See: *Xia et al. (2021). "The Causal-Neural Connection."*



## Poll 1: Understanding the SEM

### Poll Question

In the hospital messaging SEM, which of the following is the **crucial restriction** imposed by the DAG?

- (a) The structural equation functions  $f_i$  must be linear.
- (b) Each function  $f_i$  can only take the **parents** of node  $i$  as inputs.
- (c) The noise terms  $\varepsilon_i$  must be Gaussian.
- (d) The latent factors  $U_p, U_c$  must be correlated.

## **Part III: Interventions & Potential Outcomes**

What Does “Effect of Sending a Message” Really Mean?



## What Is the “Effect of Sending a Message”?

Suppose we want to measure the **effect of sending a message** (any message).

In the language of potential outcomes:

$$Y(1) = \text{outcome had we } \textit{sent} \text{ a message} \qquad Y(0) = \text{outcome had we } \textit{not}.$$

But isn't this a bit too abstract? **How do we map this to the DGP we just described?**

### A New Interpretation

Potential outcomes are the results of **interventions**: suppose we *intervened* and **forced** physicians to send a message (or not). Then:

- ▶  $Y(1)$ : the **interventional outcome** when we force  $D = 1$
- ▶  $Y(0)$ : the **interventional outcome** when we force  $D = 0$

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)]$$

# Formalizing Interventions in the SEM

What does it **formally mean** to intervene? We go to the treatment variable  $D$  and instead of its natural value, we **fix** it to some value  $d \in \{0, 1\}$ .

## DGP Under Intervention $\text{fix}(D = d)$

1. Draw  $U_p, U_c, \varepsilon_X, \varepsilon_D, \varepsilon_M, \varepsilon_Y$  from **same distribution** as the factual world.
2. Generate  $X \leftarrow f_X(U_p, \varepsilon_X)$  *(unchanged)*
3. Generate  $D \leftarrow f_D(X, U_c, \varepsilon_D)$  *(natural value, still generated)*
4. But use the **fixed** value  $d$  downstream:
5. Generate  $M(d) \leftarrow f_M(\mathbf{d}, X, U_c, \varepsilon_M)$
6. Generate  $Y(d) \leftarrow f_Y(M(d), X, U_p, \varepsilon_Y)$

**Key:** We still generate  $D$  naturally (we need it in the SWIG), but every downstream equation uses the **fixed** value  $d$  instead.

# Key Insights from the Interventional DGP

1. **Formal definition of POs:** We have given a formal mathematical definition of  $Y(1)$  and  $Y(0)$  using the *exact same building blocks* as the observed world.
2. **Consistency property:** If we set  $d = D$ , we recover the factual world:

$$Y = Y(D)$$

3. **Invariance:** The only things that remain invariant across interventions are the distribution of the noise terms/latent factors and the structural equations.

## Bottom Line

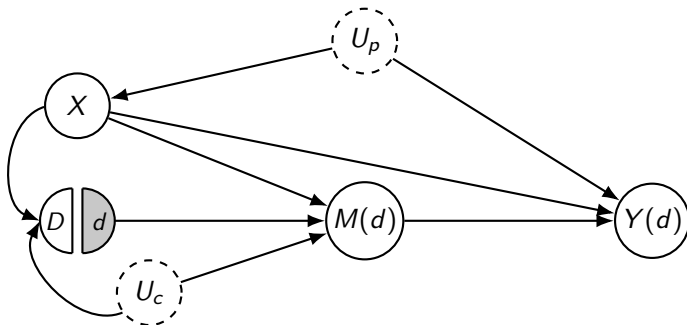
We have given an **interventional foundation** to the potential outcomes!

# The Single World Intervention Graph (SWIG)

The interventional world under  $\text{fix}(D = d)$  can be depicted by a new Bayesian network:

## SWIG Construction: “Strike Lightning on Node $D$ ”

Split  $D$  into two halves: **(i)** left semi-circle  $D$  (original) receives all **incoming** edges; **(ii)** right semi-circle  $d$  (fixed, shaded) sends all **outgoing** edges. Every variable **downstream** of the fixed value becomes an interventional random variable.



# Deducing Conditional Exogeneity from the SWIG

The SWIG is a **Bayesian network** for the interventional world. It contains both:

- ▶  $D$  — the natural treatment that *would have been* assigned
- ▶  $Y(d)$  — the counterfactual outcome

So we can check **conditional exogeneity** by checking conditional independence in this Bayesian network!

Result: Conditional Exogeneity Holds!

In the SWIG, conditioning on  $X$  blocks all paths between  $D$  and  $Y(d)$ :

$$Y(d) \perp\!\!\!\perp D \mid X$$

**Voilà!** We have deduced conditional exogeneity as a **conclusion**, where the only assumption was the **human-interpretable DAG**.

## Poll 2: What If We Don't Condition on $X$ ?

### Poll Question

In the hospital messaging SWIG, if we do **not** condition on  $X$ , is  $Y(d) \perp\!\!\!\perp D$ ?

- (a) Yes,  $Y(d)$  and  $D$  are always unconditionally independent.
- (b) No, there are open paths between  $D$  and  $Y(d)$ .

# Intervening on Multiple Variables

We can think of other interventions. For instance, intervene on the **message** too:

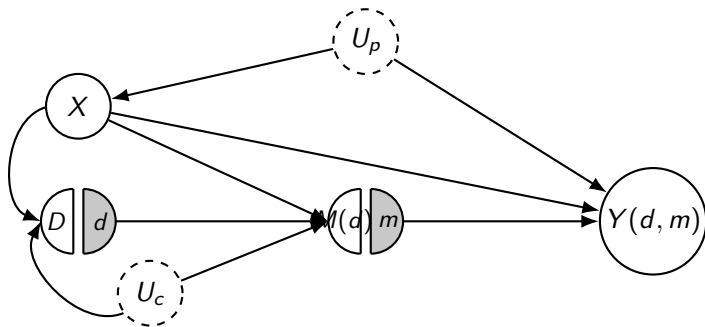
DGP Under  $\text{fix}(D = d, M = m)$

1. Draw  $U_p, U_c, \varepsilon_X, \varepsilon_D, \varepsilon_M, \varepsilon_Y$  (same distribution).
2.  $X \leftarrow f_X(U_p, \varepsilon_X)$
3.  $D \leftarrow f_D(X, U_c, \varepsilon_D)$  (natural value)
4.  $M(d) \leftarrow f_M(\textcolor{red}{d}, X, U_c, \varepsilon_M)$  (use fixed  $d$ )
5. But instead of  $M(d)$ , use the **fixed message**  $m$ :
6.  $Y(d, m) \leftarrow f_Y(\textcolor{red}{m}, X, U_p, \varepsilon_Y)$

**Notice:**  $Y(d, m) = Y(m)$  since  $d$  doesn't directly enter  $f_Y$ .

## SWIG for Double Intervention

Now we are “striking two lightnings” — intervening on both  $D$  and  $M$ :



We can check on this SWIG:  $Y(d, m) \perp\!\!\!\perp (D, M) \mid X \quad \checkmark$

*Conditional exogeneity holds again — we can apply all identification and estimation strategies from prior lectures.*



## Technical Parenthesis: Primitive Potential Outcomes

We can push interventions to the extreme — fix **all** variables, even the latent factors:

### Primitive Interventional Variables

$$X(u_p) \leftarrow f_X(u_p, \varepsilon_X)$$

$$D(x, u_c) \leftarrow f_D(x, u_c, \varepsilon_D)$$

$$M(d, x, u_c) \leftarrow f_M(d, x, u_c, \varepsilon_M)$$

$$Y(m, x, u_p) \leftarrow f_Y(m, x, u_p, \varepsilon_Y)$$

These are the **primitive potential outcomes** — the counterfactual outcome of a variable when we fix *all* of its parent variables.

Because the noise terms are independent, these primitive POs are **jointly independent**.

## Alternative Foundation: POs from Primitives

An equivalent way to define the whole framework:

**Step 1:** Nature generates all primitive POs from some exogenous distribution satisfying joint independence:

$$\{Y(m, x, u_p)\}, \{M(d, x, u_c)\}, \{D(x, u_c)\}, \{X(u_p)\}, U_p, U_c$$

**Step 2:** Then generate the observed data by “plugging in”:

$$X \leftarrow X(U_p), \quad D \leftarrow D(X, U_c), \quad M \leftarrow M(D, X, U_c), \quad Y \leftarrow Y(M, X, U_p)$$

### The Two Sides of the Same Coin

The SEM framework and the potential outcomes framework are equivalent formulations. The SEM gives a **mechanistic** foundation; the PO framework gives a **counterfactual** foundation.

# What We've Achieved So Far

Through the hospital messaging example, we have:

1. **Started** from interpretable domain assumptions → **DAG**
2. **Formalized** what the DAG means mathematically → **Non-Parametric SEM**
3. **Defined** what an intervention is and the DGP under intervention
4. **Given** an interventional interpretation of potential outcomes and ATE
5. **Used** the SWIG to visually deduce conditional exogeneity

The **only assumption** was the human-interpretable DAG. Conditional exogeneity was a **conclusion**, not a premise!

# The General Framework: DAGs and Non-Parametric SEMs

Let us now state the general definitions beyond the example.

## Definition: Non-Parametric Structural Equation Model

A **non-parametric SEM** consists of:

- (i) A set of **endogenous variables**  $V = \{V_1, \dots, V_p\}$  (observed or latent).
- (ii) A set of **exogenous noise terms**  $\{\varepsilon_1, \dots, \varepsilon_p\}$ , mutually independent.
- (iii) A set of **structural equations**:  $V_i = f_i(\text{Pa}(V_i), \varepsilon_i)$ , where  $\text{Pa}(V_i) \subset V$  are the **parents** of  $V_i$ .

The parent structure must be **acyclic**: no variable is its own ancestor.

## Definition: Directed Acyclic Graph (DAG)

The **DAG** associated with the SEM has one node per variable in  $V$  and a directed edge  $V_j \rightarrow V_i$  whenever  $V_j \in \text{Pa}(V_i)$ . Acyclicity ensures **no directed cycles**.

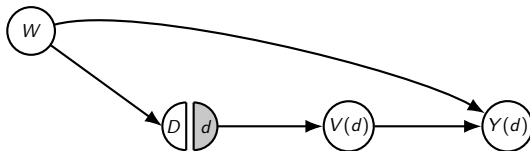
# The General Framework: Interventions and SWIGs

## Definition: Intervention $\text{fix}(D = d)$

Given a SEM and treatment  $D \in V$ , the **intervention**  $\text{fix}(D = d)$  produces a new DGP: draw all exogenous noise from the **same distribution**; generate each  $V_i$  via  $f_i$ , **except** wherever  $D$  appears as a parent, replace it with  $d$ . Downstream variables become **interventional**:  $V_i(d)$ .

## Definition: Single World Intervention Graph (SWIG)

The **SWIG** for  $\text{fix}(D = d)$ : **(1)** Split node  $D$  into two semi-circles — left half  $D$  (incoming) and right half  $d$  (shaded, outgoing). **(2)** Relabel downstream variables  $V_i$  as  $V_i(d)$ .

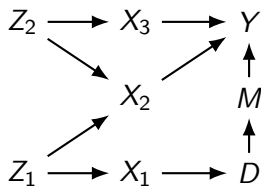


## **Part IV: In-Class Activity**

Good and Bad Controls — A Gallery of Examples

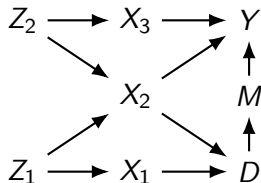
# Activity 1

Consider the following DAG:



**Task (5 min):** What are valid control sets (adjustment sets) for the causal effect of  $D$  on  $Y$ ?

## Activity 2



**Task (5 min):** What are valid control sets now?



# The General Framework: d-Separation

How do we formally check conditional independence in a DAG (or SWIG)?

## Definition: d-Separation

Let  $\mathcal{G}$  be a DAG and  $X, Y, Z$  disjoint node sets.  $X$  and  $Y$  are **d-separated** by  $Z$  ( $X \perp_{\mathcal{G}} Y \mid Z$ ) if **every path** between  $X$  and  $Y$  is **blocked** by  $Z$ . A path is blocked if it contains a node  $V$  such that:

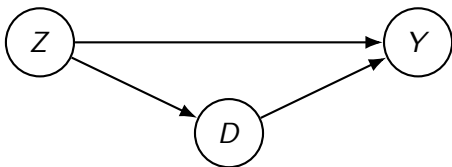
- (1)  $V$  is a **non-collider** ( $\rightarrow V \rightarrow$  or  $\leftarrow V \leftarrow$ ) and  $V \in Z$ ; **or**
- (2)  $V$  is a **collider** ( $\rightarrow V \leftarrow$ ) and neither  $V$  nor any descendant of  $V$  is in  $Z$ .

## How to Use d-Separation for Causal Inference

- (1) Construct the **SWIG** for  $\text{fix}(D = d)$ .
- (2) Check whether  $D$  and  $Y(d)$  are **d-separated** by a candidate set  $S$ .
- (3) If yes:  $Y(d) \perp\!\!\!\perp D \mid S \Rightarrow$  conditional exogeneity holds and  $S$  is a **valid adjustment set**.

## Activity 3: Good Control — Observed Common Cause

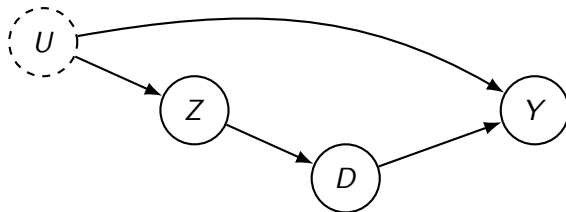
**Scenario:** A tech company wants to estimate the effect of a **coding bootcamp** ( $D$ ) on **salary** ( $Y$ ). Prior math ability ( $Z$ ) influences both who enrolls and salary.



**Good control.**  $Z$  is a common cause of  $D$  and  $Y$ . Conditioning on  $Z$  blocks the backdoor path  $D \leftarrow Z \rightarrow Y$ . (Fig. 11.3a)

## Activity 4: Good Control — Complete Treatment Proxy

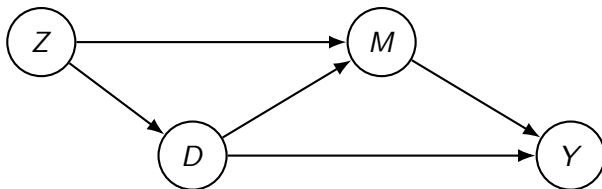
**Scenario (from textbook):** Effect of prenatal **multivitamin consumption** ( $D$ ) on **birth defects** ( $Y$ ). Unobserved genetic factors ( $U$ ) affect both. Family medical history ( $Z$ ) fully mediates  $U$ 's effect on  $D$ .



**Good control.**  $Z$  is a complete treatment proxy: it captures *all* information flow from  $U$  to  $D$ . Conditioning on  $Z$  blocks  $D \leftarrow Z \leftarrow U \rightarrow Y$ . (Fig. 11.3b)

## Activity 5: Good Control — Confounded Mediator's Common Cause

**Scenario:** Effect of **exercise program** ( $D$ ) on **cardiovascular health** ( $Y$ ), partly through diet changes ( $M$ ). Socioeconomic status ( $Z$ ) is a common cause of enrollment and diet.

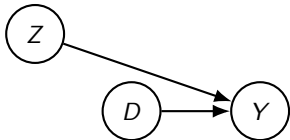


**Good control.** Without conditioning on  $Z$ , the backdoor path  $D \leftarrow Z \rightarrow M \rightarrow Y$  is open. Conditioning on  $Z$  blocks it. *(Fig. 11.4a)*

## Activity 6: Neutral Controls

### 6a: Outcome-Only Cause

*Scenario:* Effect of **tutoring** ( $D$ ) on **test score** ( $Y$ ). Student's IQ ( $Z$ ) affects only  $Y$ .



Neutral / Good

No bias. Can **improve precision** by explaining variation in  $Y$ . (Fig. 11.5a)

### 6b: Treatment-Only Cause

*Scenario:* Effect of **tutoring** ( $D$ ) on **test score** ( $Y$ ). Distance to tutor ( $Z$ ) affects only  $D$ .

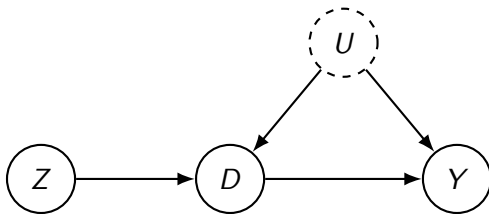


Neutral / Risky

No bias. But can **decrease precision** by explaining away useful variation in  $D$ . (Fig. 11.5b)

## Activity 7: Bad Control — Bias Amplification (Instrument)

**Scenario:** Effect of **military service** ( $D$ ) on **lifetime earnings** ( $Y$ ). Draft lottery number ( $Z$ ) affects only  $D$ . Unobserved family background ( $U$ ) confounds  $D$  and  $Y$ .

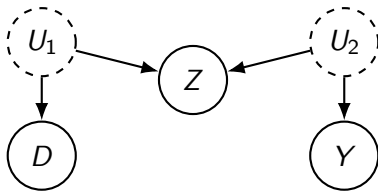


### Bad control!

$Z$  is an **instrument**. Adjusting for  $Z$  removes the exogenous variation in  $D$ , leaving only the confounded variation driven by  $U$ . This **amplifies** the bias! (Fig. 11.6)

## Activity 8: Bad Control — M-Bias (The Collider Trap)

**Scenario (from textbook):** Estimating **peer effects** on civic engagement.  $D$  = engagement of friend 1,  $Y$  = engagement of friend 2,  $Z$  = friendship. Unobserved traits ( $U_1, U_2$ ) drive both engagement and friendship (homophily).

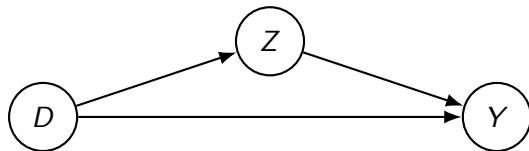


### Bad control!

$Z$  is a **collider** on  $D \leftarrow U_1 \rightarrow Z \leftarrow U_2 \rightarrow Y$ . Without conditioning: path is **blocked** at the collider. Conditioning on  $Z$  **opens** this path, creating spurious association! (Fig. 11.7)

## Activity 9: Bad Control — Mediation (Over-Control Bias)

**Scenario:** Effect of **job training program** ( $D$ ) on **wages** ( $Y$ ). The program works by improving skills ( $Z$ ), which then increase wages.



**Task:** Should we condition on  $Z$  (skills) when estimating the *total* effect?

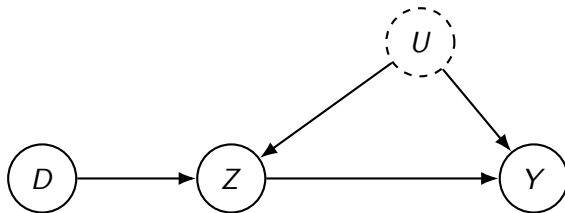
Bad control for total effect!

$Z$  is a **mediator**. Conditioning blocks the causal path  $D \rightarrow Z \rightarrow Y$ , so we only measure the *direct* effect, not the total effect. (Fig. 11.9a)



## Activity 10: Bad Control — Confounded Mediator

**Scenario:** Effect of a **new drug** ( $D$ ) on **survival** ( $Y$ ). The drug works through a biomarker ( $Z$ ). An unobserved genetic factor ( $U$ ) affects both the biomarker and survival.

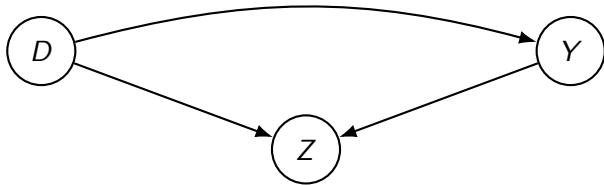


Bad control even for direct effect!

Conditioning on  $Z$  opens the collider path  $D \rightarrow Z \leftarrow U \rightarrow Y$ , introducing spurious association. *(Fig. 11.11)*

## Activity 11: Bad Control — Collider / Selection Bias

**Scenario (from textbook):** The **low birthweight paradox**. Effect of maternal **smoking** ( $D$ ) on **infant mortality** ( $Y$ ). Low birthweight ( $Z$ ) is caused by both.

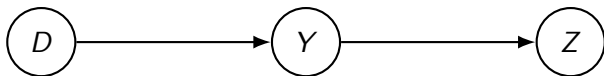


### Bad control!

$Z$  is a **collider** ( $D \rightarrow Z \leftarrow Y$ ). Conditioning on  $Z$  opens a spurious path, making smoking appear *protective* among low-birthweight babies. Classic **selection bias**. (Fig. 11.12a)

## Activity 12: Bad Control — Outcome of the Outcome

**Scenario (from textbook):** The **Industrial Growth Puzzle**. Effect of **nutrition quality** ( $D$ ) on **height** ( $Y$ ) in 19th century US/UK. Data comes from army enlistees ( $Z$ ), but enlistment is influenced by height.



### Bad control!

$Z$  is an **outcome of the outcome**. Conditioning on  $Z$  (observing only enlistees) distorts the estimated effect. Taller men had more civilian opportunities  $\Rightarrow$  selection into army is driven by height. *(Fig. 11.12c)*

## Summary: A Taxonomy of Good and Bad Controls

| Type                            | Structure                                    | Verdict | Fig.   |
|---------------------------------|----------------------------------------------|---------|--------|
| <i>Pre-treatment variables</i>  |                                              |         |        |
| Observed common cause           | $D \leftarrow Z \rightarrow Y$               | Good    | 11.3a  |
| Complete proxy of confounder    | $D \leftarrow Z \leftarrow U \rightarrow Y$  | Good    | 11.3b  |
| Confounded mediator's cause     | $D \leftarrow Z \rightarrow M \rightarrow Y$ | Good    | 11.4a  |
| Outcome-only cause              | $Z \rightarrow Y; D \rightarrow Y$           | Neutral | 11.5a  |
| Treatment-only cause            | $Z \rightarrow D \rightarrow Y$              | Risky   | 11.5b  |
| Instrument (+ unobs. conf.)     | $Z \rightarrow D; U \rightarrow D, Y$        | Bad     | 11.6   |
| M-bias (collider)               | $U_1 \rightarrow Z \leftarrow U_2$           | Bad     | 11.7   |
| <i>Post-treatment variables</i> |                                              |         |        |
| Mediator (over-control)         | $D \rightarrow Z \rightarrow Y$              | Bad     | 11.9a  |
| Confounded mediator             | $D \rightarrow Z; U \rightarrow Z, Y$        | Bad     | 11.11  |
| Collider / selection            | $D \rightarrow Z \leftarrow Y$               | Bad     | 11.12a |
| Outcome of outcome              | $D \rightarrow Y \rightarrow Z$              | Bad     | 11.12c |

# Key Lesson: Not All Controls Are Good Controls!

It matters what kind of node  $V$  is on a path:

| Node Type           | Structure                   | Conditioning           |
|---------------------|-----------------------------|------------------------|
| Collider            | $\rightarrow V \leftarrow$  | <b>Opens</b> the path  |
| Common cause (fork) | $\leftarrow V \rightarrow$  | <b>Blocks</b> the path |
| Mediator (chain)    | $\rightarrow V \rightarrow$ | <b>Blocks</b> the path |

## High-Level Rules of Thumb

- ▶ **Pre-treatment:** Adjust only for variables that are **ancestors** of  $D$ ,  $Y$ , or both. Avoid colliders and instruments (in the presence of unobserved confounding).
- ▶ **Post-treatment:** Almost always a **bad idea** to adjust. Avoid mediators, colliders, and outcomes of the outcome.

# Takeaways

1. **DAGs as Domain Assumptions:** A directed acyclic graph encodes human-interpretable assumptions about the data generating process. The associated **non-parametric SEM** formalizes it mathematically.
2. **Interventional Potential Outcomes:** Potential outcomes arise naturally as **interventional random variables** from the SEM. An intervention means “fixing” a variable to a value in the structural equations.
3. **SWIGs Deduce Conditional Exogeneity:** The Single World Intervention Graph provides a visual tool to **deduce** conditional exogeneity from domain assumptions, rather than simply assuming it.
4. **Good vs. Bad Controls:** Not all pre-treatment variables are good controls — conditioning on a **collider** can introduce bias (M-bias). Post-treatment controls are almost always bad.